University of Cape Coast

Department of Mathematics & Statistics

End of First Semester Examination (2009/2010)

MAT 101: Algebra and Trigonometry

2 HOURS

Answer all questions in Section A, which carries 50% of the total marks, and any two questions from Section B.

SECTION A

Answer all questions in this section.

1. Which of the following statements about matrix operation not true?
   A. Matrix addition is commutative.
   B. Matrix multiplication is commutative.
   C. Matrix multiplication is associative.
   D. Matrix addition is associative.

2. The polynomial $9z^4y^9 + 9z^9y^7 + 4y^9$ is a trinomial in two variables with degree..... .
   A.9 B.13 C.27 D.16

3. Let $a_n = (-1)^n n^2$, find S_4.
   A.16 B.10 C.11 D.13

4. Find the truth set of x if $\log_2(3x^2 + 8x - 1) = 1$.
   A. $\{x : x = -3, -\frac{1}{3}\}$ B. $\{x : x = 3, \frac{1}{3}\}$
   C. $\{x : x = -3, \frac{1}{3}\}$ D. $\{x : x = 3, -\frac{1}{3}\}$

5. Evaluate the determinant of the matrix $A = \begin{bmatrix} 3 & -2 & 0 \\ 2 & 5 & 8 \\ 5 & 3 & 5 \end{bmatrix}$.
   A.-97 B.63 C. 3 D. -57

6. Solve $|4 - x| < -3$
   A. $\{x > 1\}$ B. $\{x > 7\}$ C. Φ D. $\{1 < x < 7\}$

7. Evaluate $\tan(\frac{\pi}{2} - \theta)$.
   A. $Cot\,\theta$ B. $-\tan\theta$ C. $\tan\frac{\pi}{2}$ D. $Cot\frac{\pi}{2}$

8. Find the vertical asymptote (if it exist) for the curve whose equation is given as $y = \frac{x^2 + x - 6}{x + 3}$.
   A. The vertical asymptote is the line $x = -3$
   B. The vertical asymptote is the line $x = 2$
   C. The vertical asymptote is the line $x = 0$
   D. The curve does not have a vertical asymptote.

9. Find the turning point of the parabola $x^2 = x - 5$.
   A. $\left(\frac{1}{2}, \frac{19}{4}\right)$ B. $\left(-\frac{1}{2}, \frac{19}{4}\right)$ C. $\left(-\frac{1}{2}, \frac{21}{4}\right)$ D. $\left(\frac{1}{2}, -\frac{21}{4}\right)$

10. Decompose $\frac{1}{(x+3)(x+2)}$ into partial fractions.
    A. $\frac{1}{x+2} + \frac{1}{x+3}$ B. $\frac{1}{x+3} - \frac{1}{x+2}$
    C. $\frac{1}{x+2} - \frac{1}{x+3}$ D. $-\frac{1}{x+2} - \frac{1}{x+3}$

11. Find the common difference for the arithmetic sequence $x - 5b,\ x - 3b,\ x - b, \ldots$
    A. $-2b$ B. $8b$ C. $2b$ D. $-8b$

12. Solve $|x+5| = |3x-4|$.

A. $x = \frac{9}{4}$ *or* $x = -\frac{1}{4}$ B. $x = \frac{9}{2}$ *or* $x = -\frac{1}{4}$

C. $x = -\frac{9}{4}$ *or* $x = \frac{9}{2}$ D. $x = \frac{9}{4}$ *or* $x = \frac{1}{4}$

13. Evaluate the determinant of the matrix $B = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{8} & -\frac{3}{4} \end{bmatrix}$.

A. $-\frac{5}{16}$ B. $-\frac{7}{16}$

C. $-\frac{1}{16}$ D. $-\frac{4}{16}$

14. Which of the following fractions is an irrational number.

A. $\frac{1}{16}$ B. $\frac{15}{101}$ C. $\frac{2}{52}$ D. $\frac{4}{32}$

15. Solve $2\sin^2\theta = 1 + 2\sin\theta$ *for* $0 \le \theta < 2\pi$

A. -0.366 B. 1.366 C. 0.366 D. -1.366

16. Find the coefficient of x^3 in the expansion of the binomial expression $(1+3x)^{-2}$.

A. -648 B. 108 C. 648 D. -108

17. Find the vertical asymptotes (if any exist) for the curve whose equation is given as $y = \frac{3x^3 + 5x^2 - 26x + 8}{x^2 + 2x - 8}$.

A. The vertical asymptotes are the lines $x = 2$ *and* $x = 4$
B. The vertical asymptotes are the lines $x = -2$ *and* $x = -4$
C. The vertical asymptotes are the lines $x = 2$ *and* $x = -4$
D. The curve does not have a vertical asymptote

18. Find the value of x if $x = \log_{10} 0.001$.

A. $x = -10$ B. $x = -3$ C. $x = -0.001$ D. $x = -2$

19. Solve the inequality $\left|\frac{3-x}{2}\right| \ge 3$.

A. $(-\infty, -3] or [9, \infty)$ B. $(-\infty, -3]$ *or* $(9, \infty)$
C. $(-\infty, -3)$ *or* $(9, \infty)$ D. $(-\infty, -3)$ *or* $[9, \infty)$

20. Solve for X in the equation $2X + A = B$.
Where $A = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 4 \\ 2 & 1 \end{bmatrix}$

A. $\begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix}$ B. $\begin{bmatrix} -4 & 6 \\ 2 & -2 \end{bmatrix}$

C. $\begin{bmatrix} -8 & 12 \\ 4 & -4 \end{bmatrix}$ D. $\begin{bmatrix} -2 & 2 \\ 2 & 4 \end{bmatrix}$

21. Find the common difference for the arithmetic sequence $x + \sqrt{3},\ x + \sqrt{12},\ x + \sqrt{27}, \ldots$

A. $\sqrt{9}$ B. $\sqrt{15}$ C. $\sqrt{3}$ D. $\sqrt{12}$

22. What is the value of x in $2^{x}+2^{-x}=2$?

A. 0 B. 1 C. 2 D. 3

23. How many terms would the expansion of $(30x+20b)^{20}$ have?

A. 20 B. 30^{20} C. 20^{20} D. 21

24. Find the value of y if $\begin{vmatrix} 2-y & y-4 \\ 5-y & y-1 \end{vmatrix} = 15$

A. $y=\frac{1}{2}$ B. $y=2$ C. $y=\frac{1}{4}$ D. $y=4$

25. Given $\tan\theta = -\frac{5}{4}$ and $Cos\theta > 0$, find $Sec\theta$.

A. $\frac{\sqrt{41}}{5}$ B. $-\frac{5}{\sqrt{41}}$ C. $\frac{\sqrt{41}}{4}$ D. $-\frac{\sqrt{41}}{5}$

26. If $\ln z = 24$, find z.

A. $z = 2.65\times10^{10}$ B. $z=-3.18$
C. $z=1.38$ D. $z=1\times10^{24}$

27. Solve the inequality $|2x-3|\le 4$.

A. $\left[-\frac{1}{2},\frac{7}{2}\right)$ B. $\left(-\frac{1}{2},\frac{7}{2}\right]$

C. $\left(-\frac{1}{2},\frac{7}{2}\right)$ D. $\left[-\frac{1}{2},\frac{7}{2}\right]$

28. Find the horizontal asymptotes of the curve whose equation is given as $y=\frac{2x^2-5x-3}{3x^2-7x-6}$.

A. $y=0$ B. $y=\frac{1}{2}$ C. $y=\frac{2}{3}$ D. $y=1$

29. Find the sum of the first three terms of the sequence with the nth term equal to 2.

A. 0 B. 4 C. 6 D. 2

30. Which quadrant should θ lie, to make the expression $Cos\theta = -\sqrt{1-Sin^2\theta}$ true if $Sin\theta$

A. 1st quadrant B. 2nd quadrant

B. 3rd quadrant D. 4th quadrant

$\frac{1}{2^x}$

$2^x+2^{-x}=2$

$2^x(2^0+$

$2^x+\frac{1}{2^x}=2$

let $2^x=m$

$m+\frac{1}{m}=2$

$2^x=1$

SECTION B

Answer two questions only from this section.
All questions carry equal marks

1. a) If A is a 3×3 matrix and I is a 3×3 identity matrix prove that $IA = A$ and $AI = A$

b) Decompose the rational function $\dfrac{2x^2-5x+4}{x^3}$ into partial fractions.

c) Prove that $Sin\theta = \dfrac{2\tan\frac{1}{2}\theta}{1+\tan^2\frac{1}{2}\theta}$.

2. a) Find the sum of the first n terms of the sequence whose n^{th} term is $2^n - 2n + 1$

b) Given that $\log_e(p+2) + \log_e q = r - \dfrac{1}{3}$

and $\log_e(p-2) - \log_e q = 2r + 1$

Show that $p^2 = 4 + 32^r$. If $r = 1$, find the possible values Of p and q.

c) Your broker tells you of two investments she thinks are worthwhile. She advises a new issue of P.Coles stock which should yield 20% over the next year, and then to balance your account she advises Cape Coast Metropolitan bonds with a 10% yearly yield. The stock to bond ratio should be no less than 3 to 1. If you have no more than 100,000 Ghana cedis to invest and do not want to invest more than 70,000 Ghana cedis in P.Coles or less than 20,000 Ghana cedis in bonds, how much should be invested in each to maximize your return.

3. a) Use binomial series expansion to approximate $(2.99)^{\frac{1}{3}}$, correct to five decimal places.

b) Solve the inequality below, expressing the solution set in interval notation.

$$\left|\frac{x+4}{x-5}\right| \le 3$$

c) Prove that $\dfrac{Sec2\theta + Cot2\theta}{Sec2\theta} = 1 + Cosec2\theta - Sin2\theta$

4. a) Solve the system of equations below

$$\begin{cases} x - 2y - 5z = -12 \\ 2x - y = 7 \\ 5y + 6z = 4x + 1 \end{cases}$$

using a matrix approach.

b) Sketch the graph of the function $y + 1 = \dfrac{-3}{x-2}$

c) Prove that $\dfrac{Sec^2 2\theta - \tan^2 2\theta}{\tan 2\theta + Sec2\theta} = \dfrac{Cos2\theta}{1+Sin2\theta}$